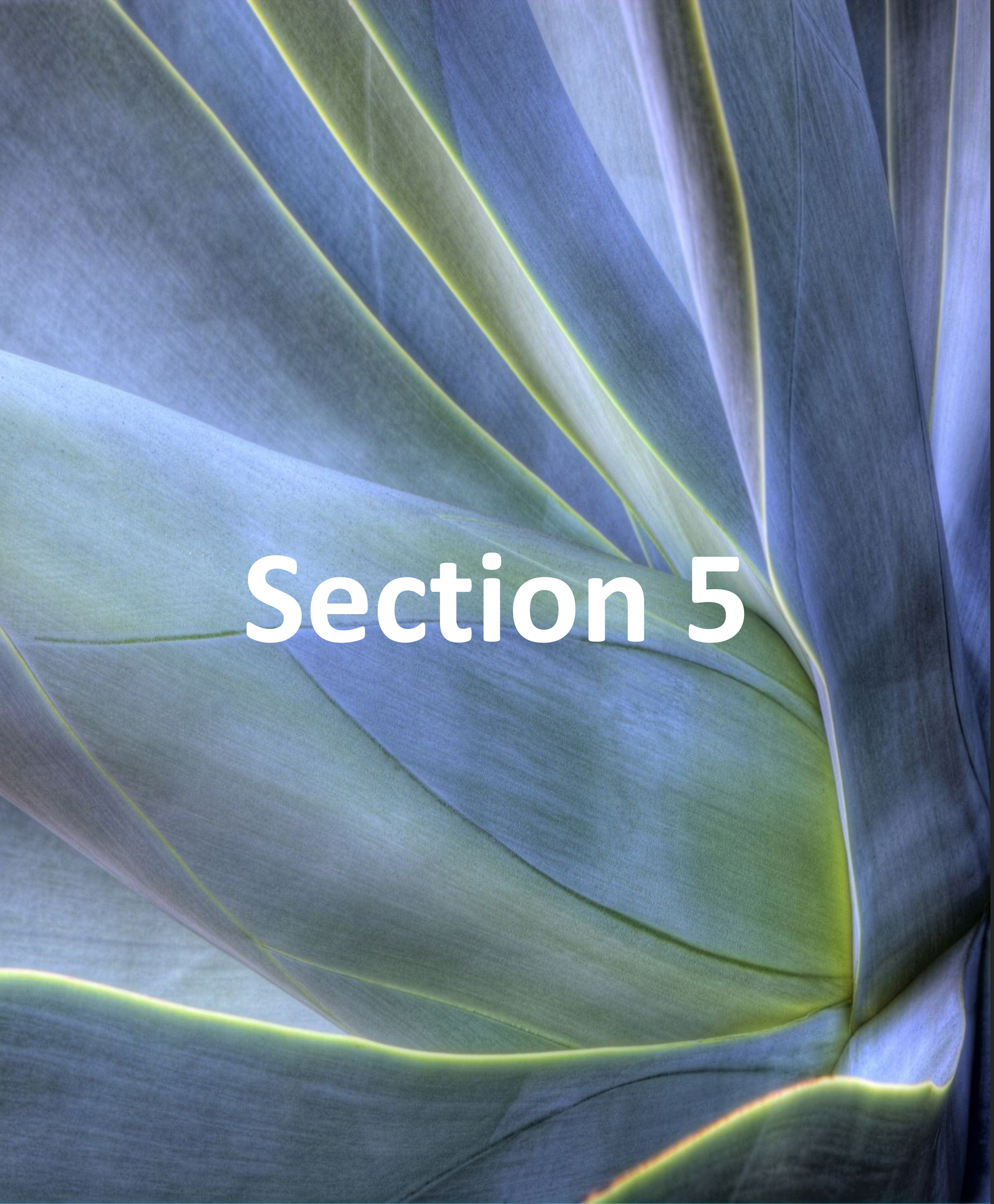
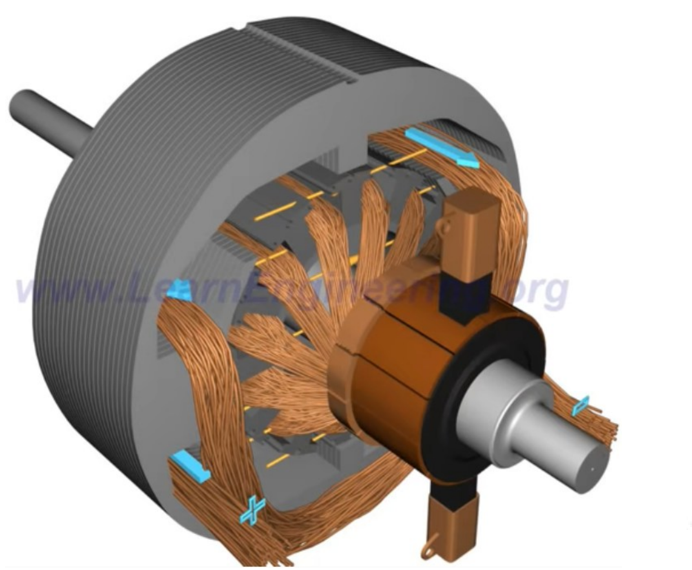


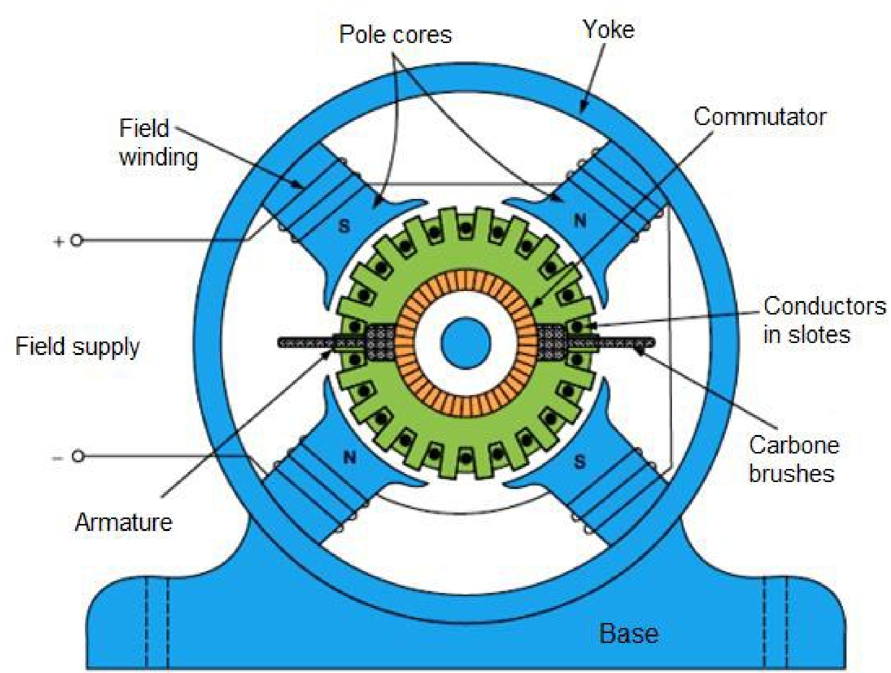
Section 5

DC Motor

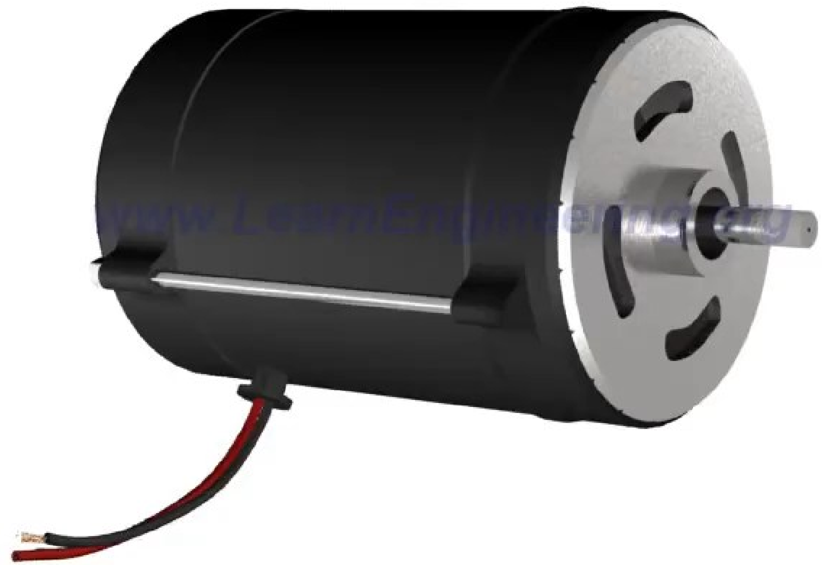
1. Introduction to DC Motor

- **DC (direct current) motor** is an electric motor that runs on direct current (DC) electricity.
- The physical structure of the machine consists of **two parts:** The **stator or stationary** part and the **rotor or rotating part**. The stationary part of the machine consists of the frame, which provides physical support, and the pole pieces, which project inward and provide a path for the magnetic flux in the machine.



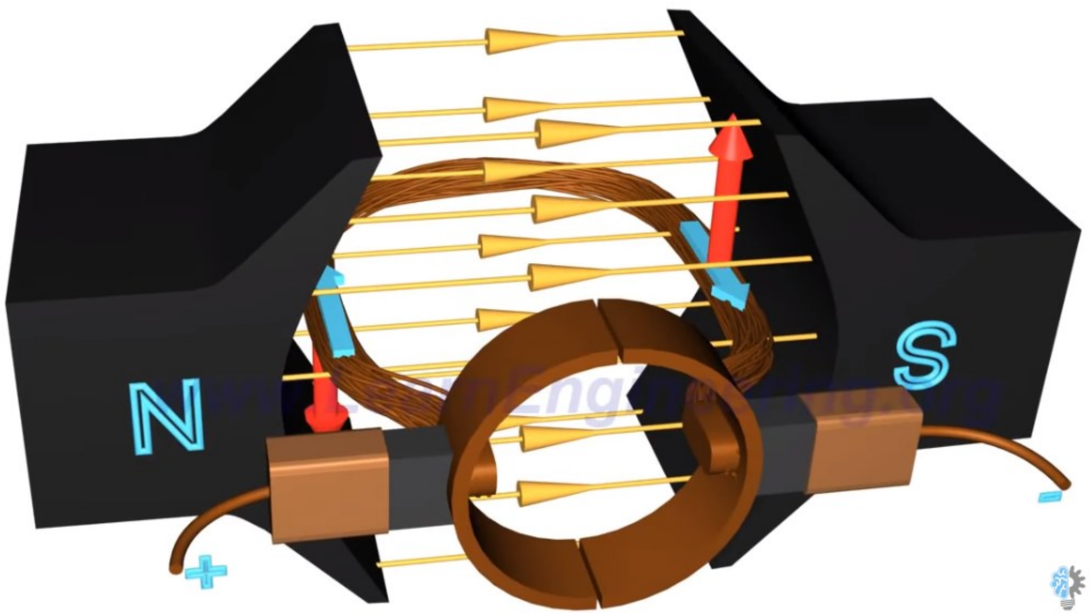


2. Working principle of DC motor

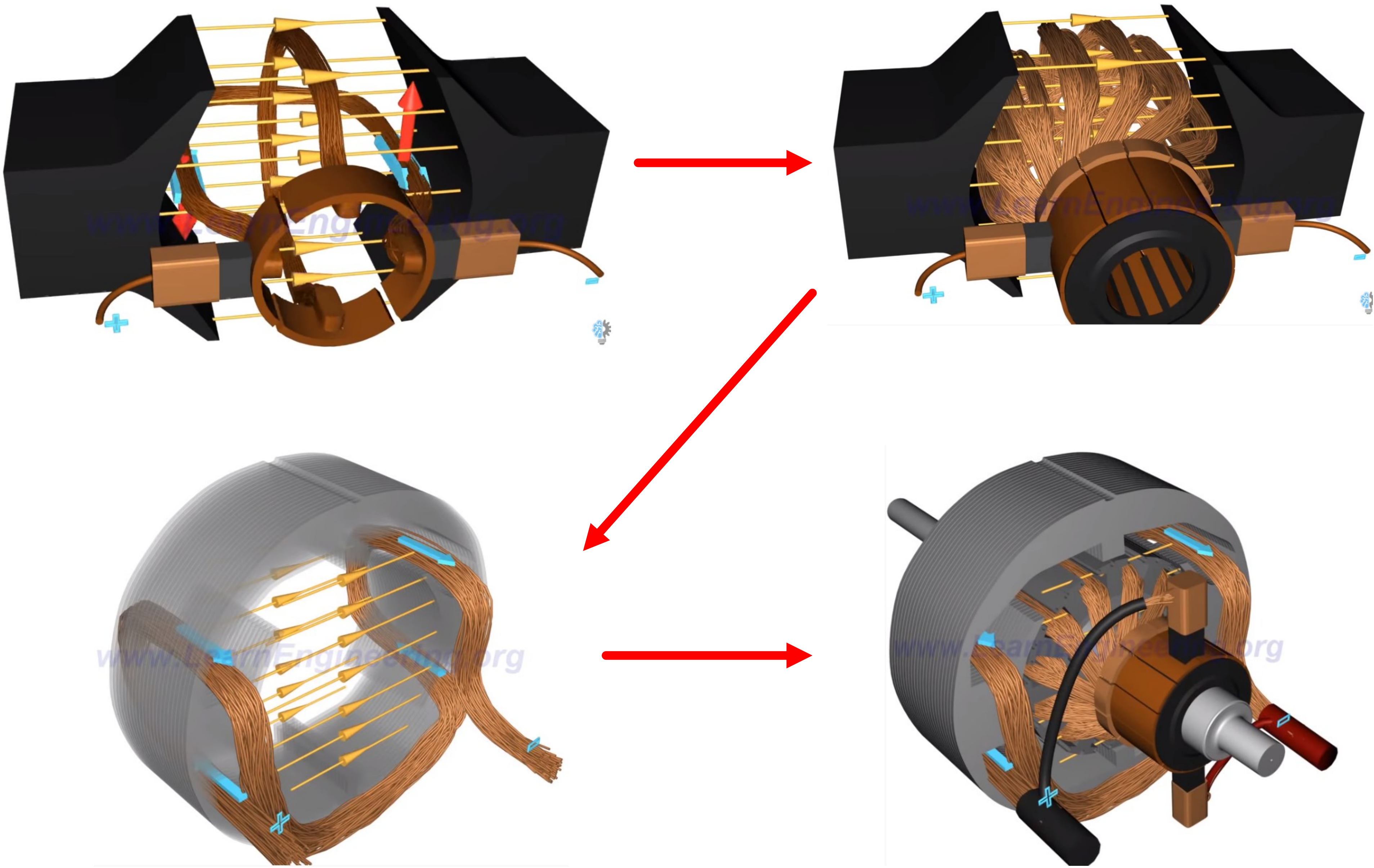


2. Working principle of DC motor

- There are two principal windings on a **DC machine**: the armature windings and the field windings.
- The armature windings are defined as the windings in which a voltage is induced, and the field windings are defined as the windings that produce the main magnetic flux in the machine.
- In a normal **DC machine**, the armature windings are located on the rotor, and the field windings are located on the stator.
  - ➢ When a current-carrying conductor is placed in an external magnetic field, it will experience a force.
  - ➢ Due to this force, torque is produced which rotates the rotor of motor and hence the motor runs.

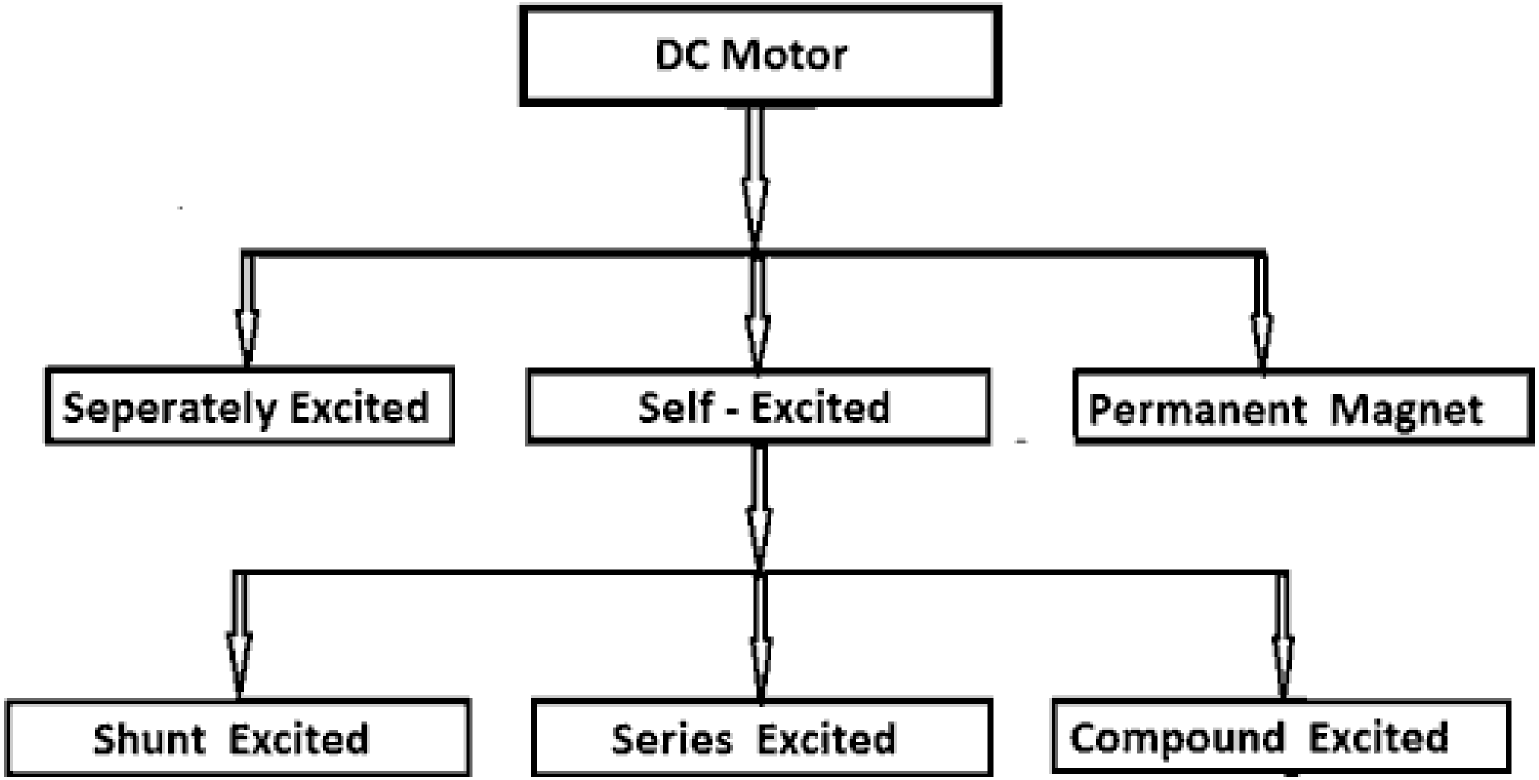


2. Working principle of DC motor



3. Type of DC motor

Types of DC motors in general use:



3. Type of DC motor

Characteristics and Applications of Different Types of DC Motor

1. Separately Excited DC Motors

Characteristics:

- Separately excited DC motors got very accurate speed.
- These type of motors are much suitable for applications which need speed variation from low speed to higher speed.

Applications

- Paper Machines
- Rolling Units
- Electric of Ships

2. Shunt DC Motors

Characteristics:

- Shunt motors are normally got constant speed.
- The starting torque of this motor is medium which is limited to nearly 200% by commutation.
- Speed control of shunt motor done by decreasing armature voltage control

3. Type of DC motor

Applications

- ➢ Fans and Blowers
- ➢ Wood working machines
- ➢ Centrifugal pumps

3. DC Series Motors

Characteristics:

Unlike DC Shunt Motors this DC Series motors got variable speed and the starting torque of this DC motor is quite high. Normally it can high up to 500%. This DC series motor speed control done by using series resistance method.

Applications

- ➢ Electrical Cranes
- ➢ Trolley Cars
- ➢ Conveyors Belt Drives
- ➢ Electric locomotives

3. Type of DC motor

4. Compound DC Motors

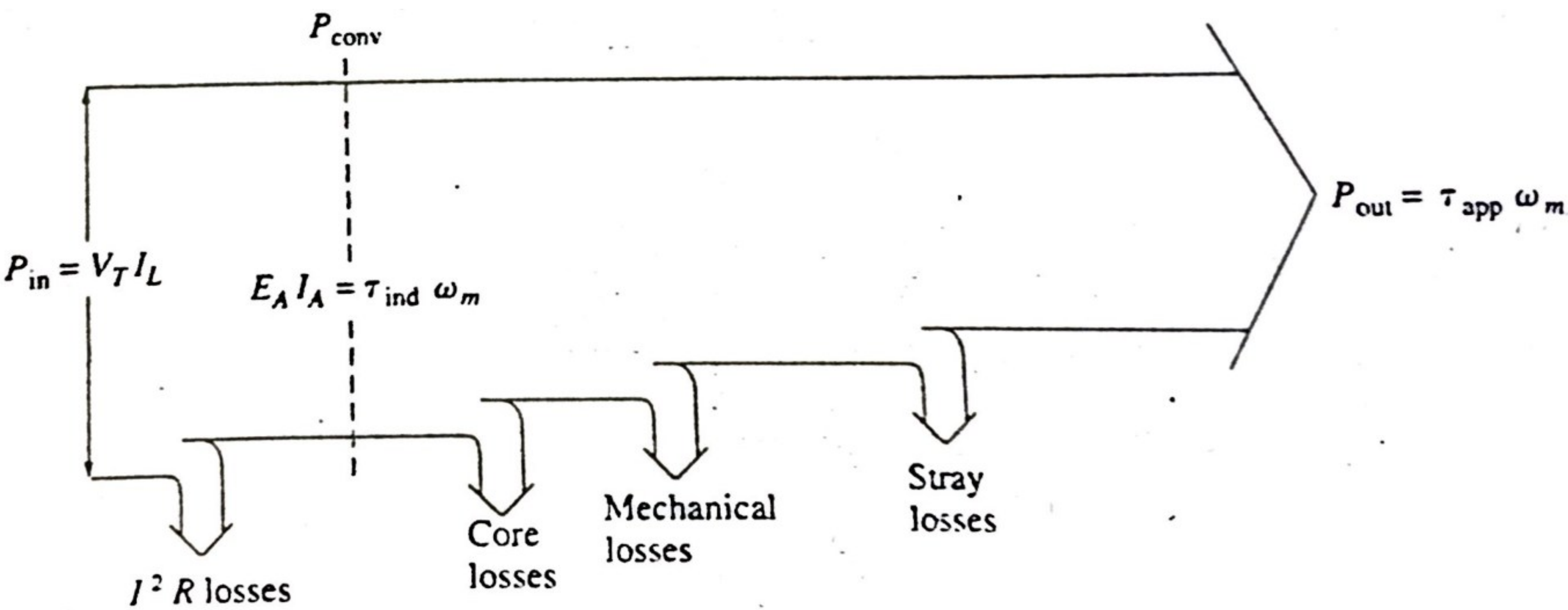
Characteristics:

- Normally these type of DC motor is good for adjustable varying speed. The speed regulation is varying 25%-30%.
- The starting torque of this DC motor is also high like DC series motors

Applications

- Electrical Elevators
- Conveyor Belt Drive
- Heavy Planes
- Rolling Mills

4. Power Flow and Losses in DC motor



DC motors take in electric power and produce mechanical power.

Efficiency: $$\eta = \frac{P_{out}}{P_{in}} \times 100\%$$

The losses in DC motor:

- Electrical or copper losses
- Brush losses
- Core losses
- Mechanical losses
- Stray load losses

4. Power Flow and Losses in DC motor

- **Electrical or copper losses:** Copper losses are the losses that occur in the armature and field windings of the machine. The copper losses for the armature and field windings are given by:

Armature loss: $P_A = I_A^2 R_A$

Field loss: $P_F = I_F^2 R_F$

where
P_A = armature loss
P_F = field circuit loss
I_A = armature current
I_F = field current
R_A = armature resistance
R_F = field resistance

Note: The resistance used in these calculations is usually the winding resistance at normal operating temperature.

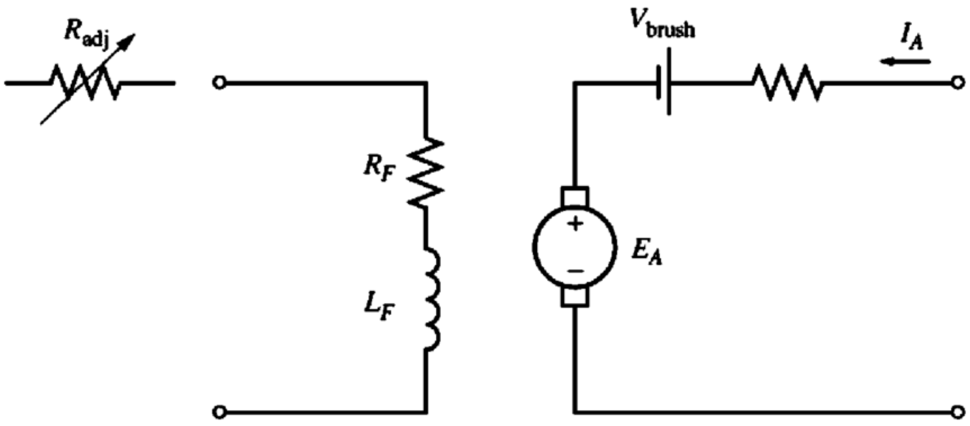
- **Brush losses:** The brush drop loss is the power lost across the contact potential at the brushes of the machine. The brush voltage drop is usually assumed to be about 2V. It is given by the equation: $P_{BD} = V_{BD} I_A$

where
P_{BD} = brush drop loss
V_{BD} = brush voltage drop
I_A = armature current

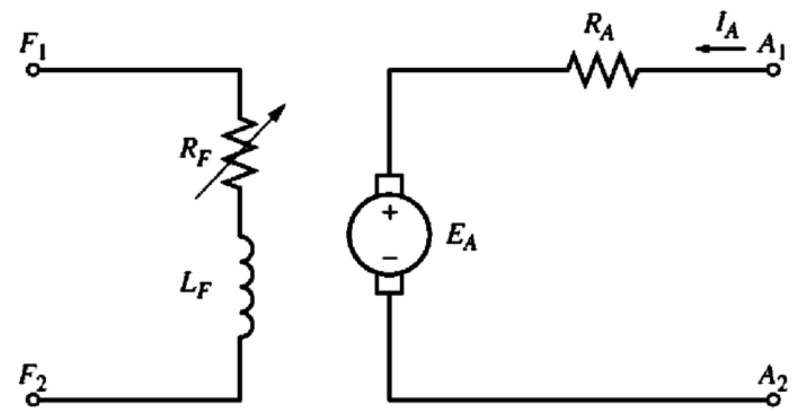
4. Power Flow and Losses in DC motor

- **Core losses:** The core losses are occurring in the metal of the motor.
- **Mechanical losses:** The mechanical losses in a DC machine are the losses associated with mechanical effects. There are two basic types of mechanical losses: friction and windage. Friction losses are losses caused by the friction of the bearings in the machine, while windage losses are caused by the friction between the moving parts of the machine and the air in side the motor 's casing.
- **Stray losses:** Stray losses are losses that cannot be placed in one of the previous categories. No matter how carefully losses are accounted for, some always escape inclusion in one of the above categories. All such losses are lumped into stray losses. For most machines, stray losses are taken by convention to be 1% of full load.

5. The equivalent circuit of a DC motor



(a)



(b)

E_A : Internal generated voltage (armature circuit)

$$\boxed{E_A = K\phi\omega}$$

$$\boxed{\tau_{ind} = K\phi I_A}$$

R_A Resistor (armature circuit)

V_{brush} : Brush voltage drop

L_F : Inductor (field coils)

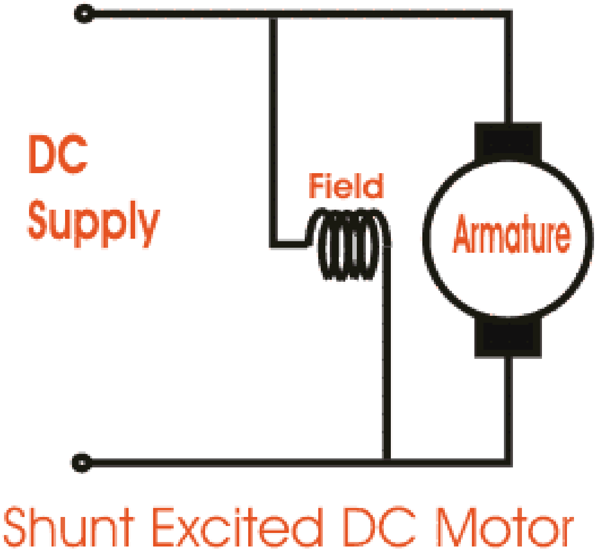
R_F : Resistor (field coils)

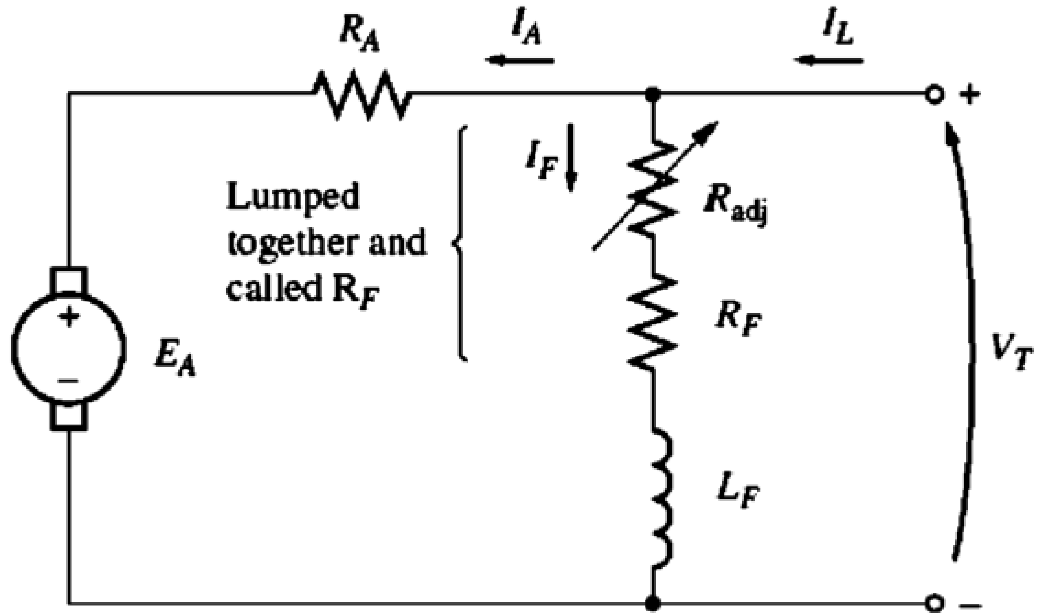
R_{adj} : External variable resistor (for field current control)

6. Self-excited DC motor

Shunt DC motor

A shunt DC motor is a motor whose field circuit gets its power directly across the armature terminal of the motor. The equivalent circuit of a separately DC motor is shown as figure below.





$$I_F = \frac{V_T}{R_F}$$

$$I_L = I_A + I_F$$

$$V_T = E_A + I_A R_A$$

6. Self-excited DC motor

Shunt DC motor

The characteristic of a Shunt DC motor

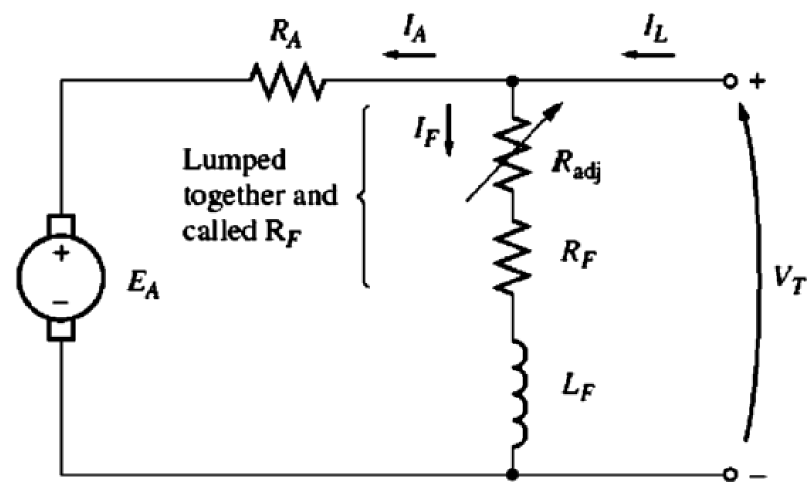
The KVL equation: $V_T = K\phi\omega + I_A R_A$

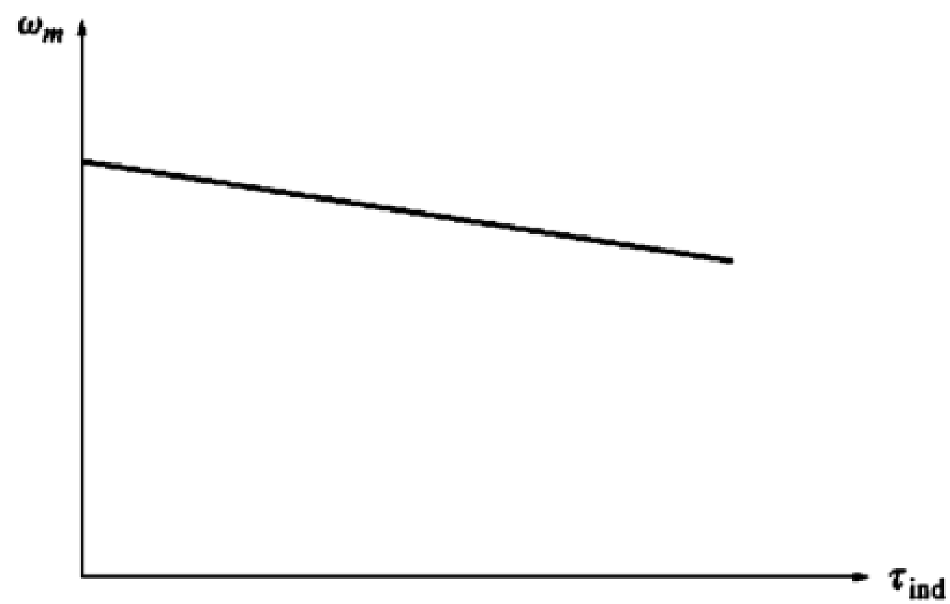
$$E_A = K\varphi\omega$$

Since: $$I_A = \frac{\tau_{ind}}{K\phi}$$

Thus: $$V_T = K\phi\omega + \frac{\tau_{ind}}{K\phi} R_A$$

Finally: $$\omega = \frac{V_T}{K\phi} - \frac{R_A}{(K\phi)^2}\tau_{ind}$$





6. Self-excited DC motor

Shunt DC motor

How does a shunt DC motor respond to a load?

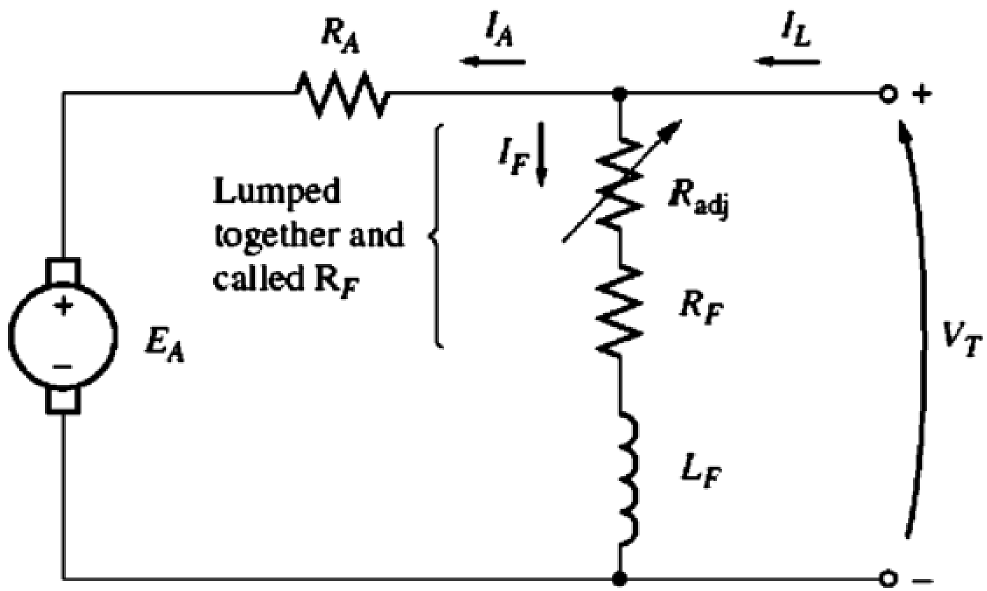
- Supposed that the load on shaft of shunt DC motor increased, thus the load torque is increased: $\tau_{load}^{\uparrow}$
- When: $\tau_{load} \succ \tau_{ind}$ the machine will start slow down
- The internal generated voltage drop: $E_A = K\varphi\omega \downarrow$
- So, the armature current in the motor is increased: $I_A = \dfrac{V_T - E_{A\downarrow}}{R_A} \uparrow$
- Since the armature current rise, the induced torque in the motor increased: $\tau_{ind} = K\varphi I_A \uparrow$
- Finally, the induced torque will equal the load torque at a lower mechanical speed of rotation.

Example 1

A shunt DC motor is supplied from $V = 220\ V$. Armature resistance $R_a = 1.0\,\Omega$. Field winding resistance $R_f = 110\,\Omega$. Constant Kϕ= 1.35Speed under load is $1200\ rpm$.

Calculate:

a. Field current I_f.
b. Internal generated voltage E_A.
c. Armature current I_a.
d. Induce torque τ_{ind}
e. Developed power P_{dev}.
f. Total electrical input



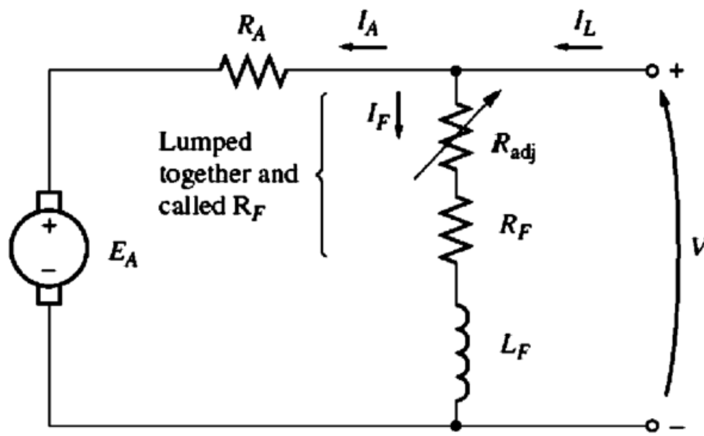
6. Self-excited DC motor

Speed Control of Shunt DC Motors

The two common ways in which the speed of a shunt dc machine can be controlled are:

1. Adjusting the field **resistance** $\boldsymbol{R_F}$
2. Adjusting the **terminal voltage** applied to the armature

Adjusting the field resistance R_F



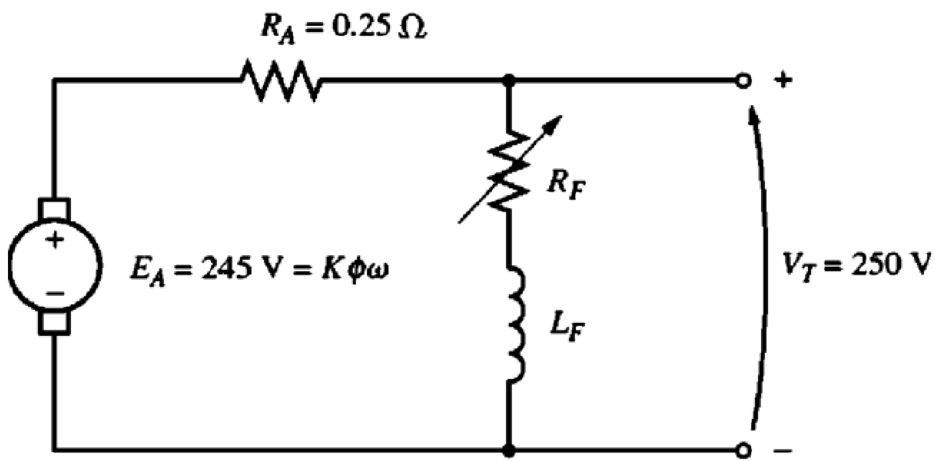
- Increasing R_F causes $I_F (= V_T/R_F \uparrow)$ to decrease
- Decreasing I_F decreases ϕ
- Decreasing ϕ lowers $E_A (= K\phi \downarrow \omega)$
- Decreasing E_A increases $I_A (= V_T - E_A \downarrow)/R_A$.
- Increasing I_A increases $\tau_{ind} (= K\phi \downarrow I_A \uparrow\uparrow)$, (with the change in I_A dominant over the change in flux)
- Increasing τ_{ind} makes $\tau_{ind} > \tau_{load}$, and the speed ω increases
- Increasing $E_A = K\phi\omega \uparrow$ again
- Increasing E_A decreases I_A.
- Decreasing I_A decreases τ_{ind} until $\tau_{ind} = \tau_{load}$ at a higher speed ω.

6. Self-excited DC motor

Speed Control of Shunt DC Motors

Adjusting the field resistance R_F (Justify)

A shunt dc motor with an internal resistance of 0.25 Ω. It is currently operating with a terminal voltage of 250V and an internal generated voltage of 245V. Therefore, the armature current flow is $I_A = \frac{(250\mathbf{V} - 245\text{ V})}{0.25\ \Omega} = 20\text{A}$



What happens in this motor if there is a 1 percent decrease in flux?

If the flux decreases by 1 percent, then E_A must decrease by 1 percent too, because $E_A = K\phi\omega$ .Therefore, E_A will drop to:

$$E_{A2} = 0.99\, E_{A1} = 0.99(245\ V) = 242.55\ V$$

6. Self-excited DC motor

Speed Control of Shunt DC Motors

Adjusting the field resistance R_{F} (Justify)

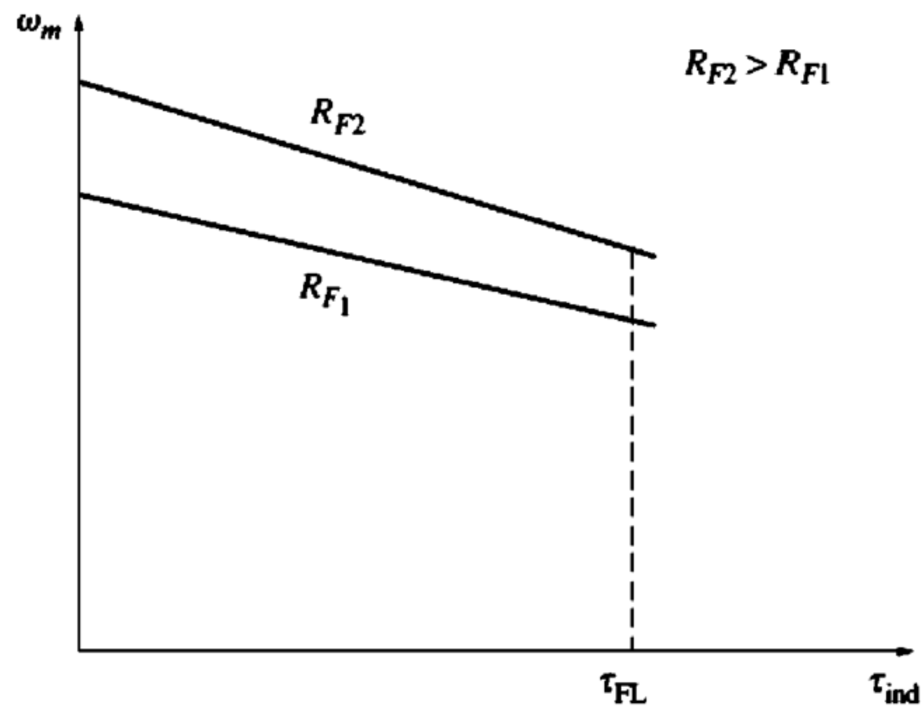
$$E_{\mathrm{A2}} = 0.99\, E_{\mathrm{A1}} = 0.99(245\, V) = 242.55\, V$$

The armature current must then rise to

$$\mathrm{I}_{\mathrm{A}} = \frac{250\, V - 242.55\, V}{0.25\, \Omega} = 29.8\, A$$

Thus, a **1 percent** decrease in flux produced a **49 percent** increase in armature current.

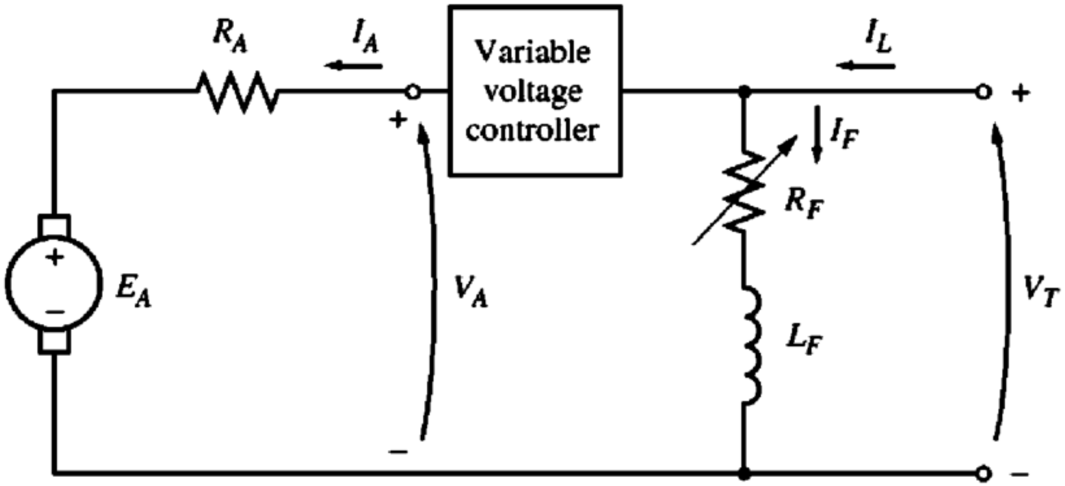
$$\tau_{\text{ind}} = K\overset{\downarrow}{\phi}\,\overset{\Uparrow}{I_A}$$

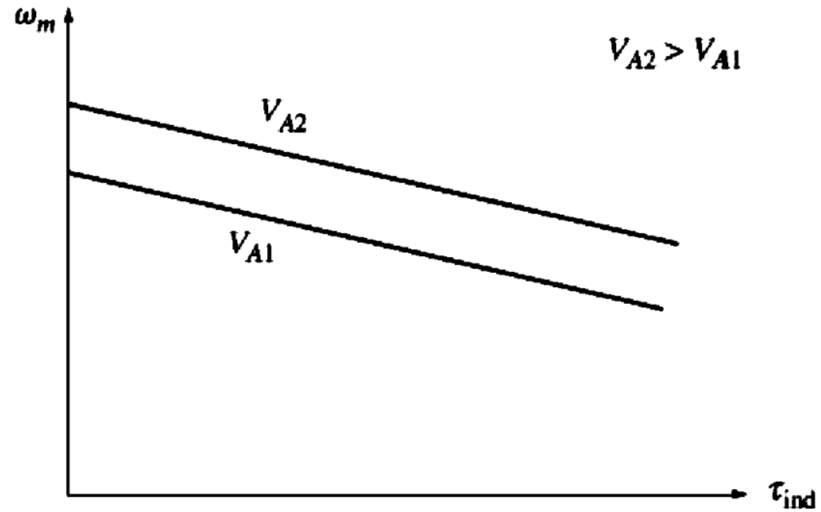


6. Self-excited DC motor

Speed Control of Shunt DC Motors

Adjusting the terminal voltage applied to the armature





V_T is constant
V_A is variable

- An increase in V_A increases $I_A = \frac{(V_A\uparrow - E_A)}{R_A}$
- Increasing I_A increases $\tau_{ind} = K\phi I_A \uparrow$
- Increasing τ_{ind} makes $\tau_{ind} > \tau_{load}$,increasing ω
- Increasing ω increases $E_A = K\phi\omega \uparrow$.
- Increasing E_A decreases $I_A = \frac{(V_A - E_A\uparrow)}{R_A}$
- Decreasing I_A decreases τ_{ind} until $\tau_{ind} = \tau_{load}$ at a higher ω.

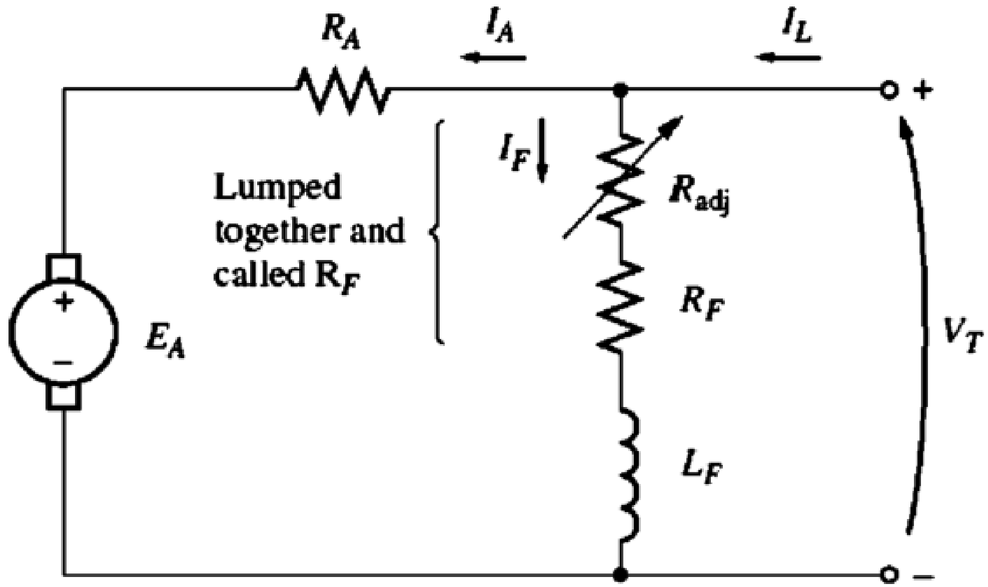
Example 2

A 220-V shunt DC motor has the following parameters:

- Armature resistance: $R_a = 0.5\Omega$
- Field resistance: $R_f = 220\Omega$
- Rated supply voltage: 220 V

The motor runs at 1200rpm and draws an armature current of 20A at rated voltage. Assume Kϕ is not change.

a. Calculate the $E_A$at rated conditions.
b. If the supply voltage is reduced to 180V, and the armature current remains 20A, calculate the new speed of the motor.



Example 3

A 50-hp, 250-V, 1200 r/min dc shunt motor with compensating windings has an armature resistance (including the brushes, compensating windings, and interpoles) of 0.06Ω. Its field circuit has a total resistance $R_{adj} + R_F$ of 50Ω, which produces a no-load speed of 1200 r/min. There are 1200 turns per pole on the shunt field winding:

a. Find the speed of this motor when its input current is 100 A.
b. Find the speed of this motor when its input current is 200 A.
c. Find the speed of this motor when its input current is 300 A.
d. Plot the torque-speed characteristic of this motor.